

VisiTALK

Communication & Emotional Support
Solution for Autistic Children



Bridging the communication gap between autistic children and
their parents through structured tracking and emotional
support.

Specification Document

Course Name: Software Engineering

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Version History

Version	Date	Description of Changes
v1.0	Mar 17, 2026	Added ElevatorPitch, Impact Mapping, Personas, and Interview Reports.
v2.0	Mar 29, 2026	Add Story Mapping, User Stories, MVP, and Release Plan



1. Project Vision And Elevator Pitch

Picture a service available to parents that allows them to communicate easily with their autistic kids as they do via WhatsApp or Instagram.

For many autistic children, communicating can be a challenge since they sometimes struggle to express themselves or their wants and needs clearly to their families. The app allows parents to have a simple matter of facial expressions and visual images to understand how to assist their child in both communication and development.

With the use of the visual card messaging system created for children, children can express their day in a visual format by sending photos to their parents and keeping their feelings in an emoji diary. If they feel overwhelmed and want to communicate this with their parent, they can simply touch the SOS button to contact their parent in seconds.

The app also provides children with a step-by-step visual schedule designed to support them in becoming independent by teaching them how to complete their daily routine and tasks, and provides parents with a weekly summary of their child's feelings and activities; thus, parents can better understand their child's progress.

The app assists families in establishing a strong connection through communication, emotional expression, and structured guidance, and assists autistic children in developing themselves daily.

2. User Research And Validations

2.1 Interviews And Survey Reports

Interview time: 05/03/2026, Thursday 3 PM



Project Objective: To design a digital solution that bridges the communication gap between autistic children and their parents, while providing structured support and emotional tracking.

Part 1: General Question

1.1 In your daily life, what are the primary ways your child communicates their needs, desires, or discomforts to you (e.g., verbal, gestures, visual cards)?

- Before 4 years old, cards with pictures. Then complete the sentence with cards. However, can not get the inferred ideas of a sentence. (e.g. dying to)

Part 2: Exploring Visual Communication & Emotional Expression (Feature 1)

Our concept includes a text-free, visual-first home screen using icons/emojis, along with an "emotional doodle pad."

2.1 Does your child ever use drawing, coloring, or scribbling as a way to self-regulate or express feelings when they are overwhelmed?

- Very hard to draw or write something. (Communication) PECS

Part 3: Managing Emotional Overload & Tracking (Feature 2)

Our concept includes a simple "SOS button" for the child to express negative emotions/sensory overload, generating a weekly data report for parents.

3.1 What are the early signs or triggers that indicate your child is approaching a meltdown or experiencing sensory overload?

- No. Ask questions many times/ Laughing/ Crying for no reason. Symptoms are different from person to person.



Maybe try another way, we can use a private note for children, and parents can get the statistics

- **3.3** From a parent's perspective, if an app could automatically track your child's emotional fluctuations, what specific data or trends would you most want to see in a weekly report (e.g., triggers, time of day, frequency)?

Weekly report

Part 4: Structured Routines & Task Execution (Feature 3)

Our concept includes replacing standard "posts" with everyday action buttons and visual task-oriented guidance (e.g., breaking down "going to the supermarket" into steps).

- **4.1** Would sending your child a "visual task card" (where they can check off simple steps like: 1. Get basket -> 2. Find milk -> 3. Go to cashier) be a practical way to encourage their independence?

Planning (Scheduling)

2.2 Key Insights And Feature Iteration

1. Key Insights from the Interview

The interview provided crucial real-world feedback that helped us correct our initial assumptions:

- **Literal Communication:** The child relies heavily on visual cards (PECS) and struggles to understand abstract concepts, metaphors, or inferred meanings.
- **Drawing is Too Difficult:** Using a "doodle pad" to express emotions is not practical. Fine motor skills and cognitive load make drawing a frustrating experience rather than a relief.



- **Individualized Triggers:** Early signs of an emotional meltdown vary greatly from person to person (e.g., asking repetitive questions, laughing, or crying for no reason). A simple "SOS button" is not enough to capture these unique patterns.

2. Refined User Personas

Based on the feedback, we updated our target user profiles:

- **The Child (Core User):** Needs direct, literal visual communication (PECS) . Requires a highly structured visual schedule to understand what happens next.
- **The Parent (Secondary User):** Needs a private space to log personalized behavioral triggers and requires automated **weekly reports** to track emotional trends over time.

3. Feature Iterations (What We Changed)

Feature 1: Visual Communication

- **Before:** Icon-based home screen + Emotional Doodle Pad.
- **After: Plus PECS Sentence Builder.**
- **Why:** We will focus on a digital PECS (Picture Exchange Communication System) board where the child can drag and drop clear, literal pictures to form simple sentences (e.g., "I want" + "Apple").

Feature 2: Emotional Tracking

- **Before:** A universal "SOS Panic Button".
- **After: Private Behavior Logger & Weekly Reports.**
- **Why:** Since meltdown signs are highly individual, parents need a "Private Note" feature to log specific behaviors (like random laughing). The system will



analyze these notes to generate the **Weekly Report** requested by the parent.

Feature 3: Task Execution

- **Before:** Single Visual Task Cards (e.g., "Go to supermarket").
- **After: Visual Scheduling & Planning.**
- **Why:** The core need is *scheduling*. We will upgrade this feature into a step-by-step daily timeline. The child can check off tasks as they complete them, encouraging independence through structured planning.

3. Target User Personas

Given the unique nature of VisiTALK, our system architecture must accommodate a highly specific **dual-user dynamic**. The application is fundamentally designed to serve two entirely distinct mental models and usability requirements simultaneously:

- **The Primary User (The Child):** Requires extreme visual simplicity, zero text dependency, and an error-forgiving interface to reduce cognitive load.
- **The Secondary User (The Parent/Caregiver):** Requires a secure environment and analytical reporting.

3.1 Primary Persona: The Autistic Child

Leo is a non-verbal or minimally verbal child who processes information linearly and visually. He interacts with the world through images and predictable patterns rather than complex verbal instructions or text.

- **Core Goals (What he wants to achieve):**



- To express his immediate needs (e.g., hunger, fatigue) quickly and clearly without being misunderstood.
- To understand exactly what is happening next in his day to feel safe and in control.
- **To discover and engage in suitable activities and environments** (such as identifying which local park to visit or what activity to play) using clear, recognizable visual cues.

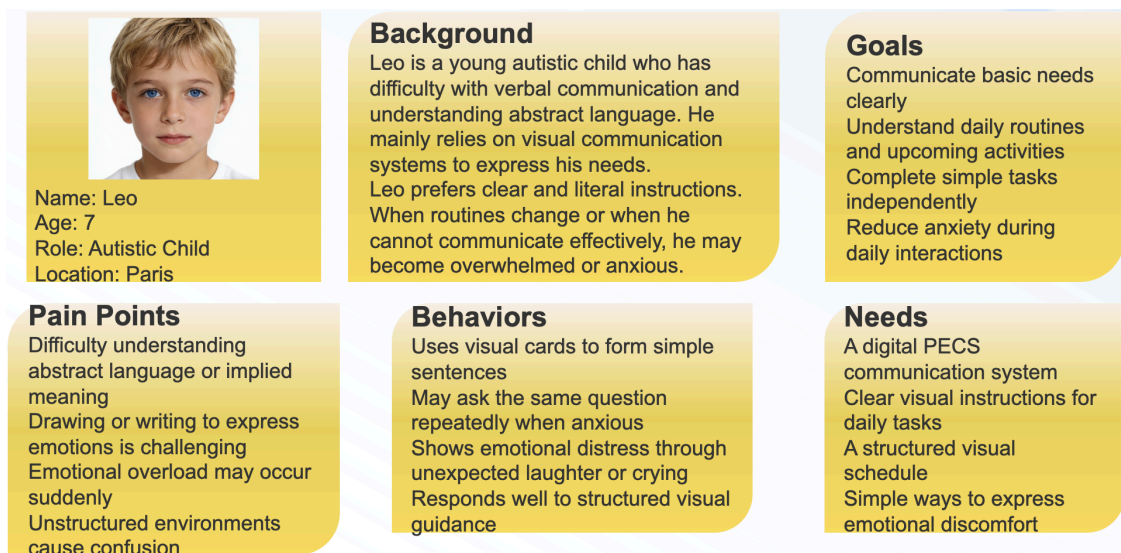


Figure 1: Primary User Persona – The Autistic Child

3.2 Secondary Persona: The Parent And Caregiver

Claire is a working mother who acts as the primary caregiver and administrative manager of Leo's schedule, therapy, and emotional well-being. She is tech-savvy but time-poor, constantly juggling work and caregiving.

- **Core Goals (What she wants to achieve):**
 - To quickly build and adjust step-by-step visual schedules for Leo's daily routines.
 - **To find, plan, and manage suitable activities and safe environments** (e.g., evaluating if a specific park



or public space is appropriate for Leo's current sensory threshold).

- To seamlessly log behavioral triggers, environmental contexts, and emotional pre-cursors in real-time without drawing Leo's attention.
- To generate objective data reports to share with Leo's therapists or pediatricians.

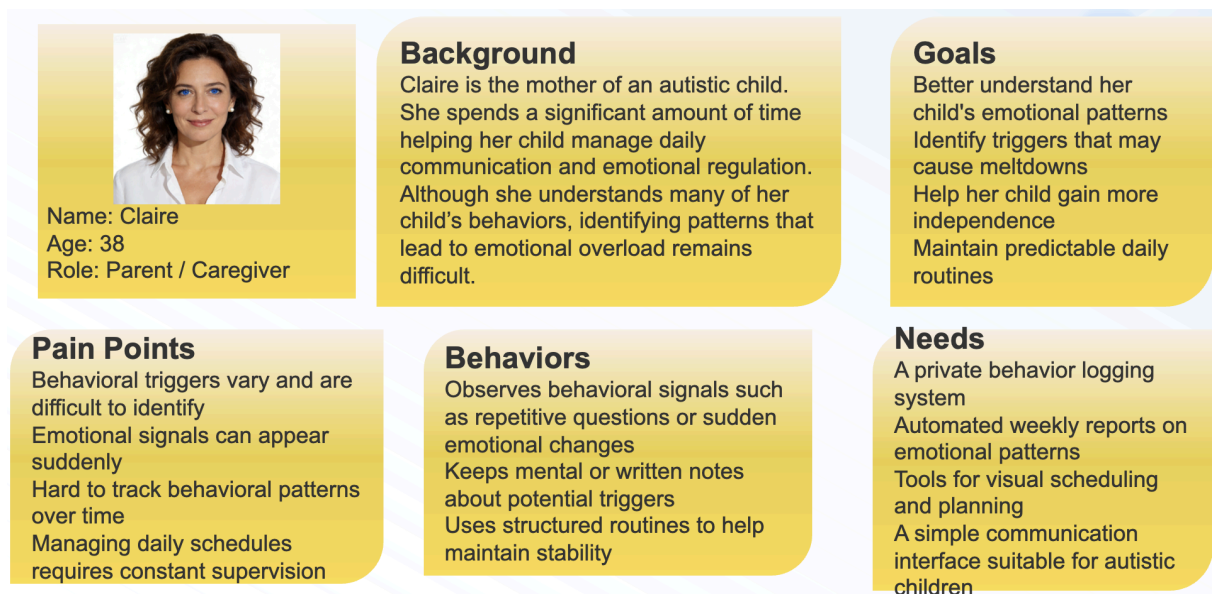


Figure 2: Secondary User Persona – The Parent

4. Strategic Alignment And Impact Mapping

4.1 Core Goals

The foundation of our strategic alignment rests on two primary objectives:

1. **Primary Goal:** To bridge the communication gap for non-verbal or minimally verbal autistic children by providing an accessible, visual-first medium for expressing immediate needs.



2. **Secondary Goal:** To reduce the frequency of emotional meltdowns by providing parents with a structured environment and data-driven insights to identify behavioral triggers.

4.2 Impact Map Breakdown

By mapping the overarching goals to the specific actors, the behavioral changes we wish to inspire, and the final software features, we defined the strict scope of our Minimum Viable Product (MVP). The logic chain is detailed in the table below:

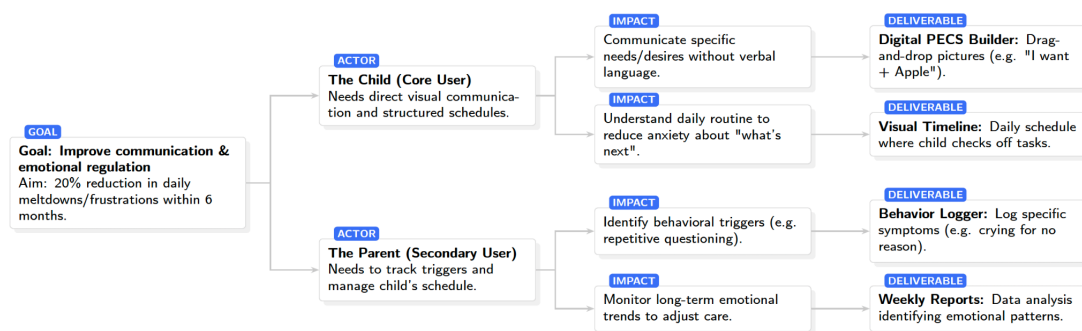


Figure 3. Strategic Impact Mapping for VisiTALK

5. Product Scope And Story Mapping

To effectively translate our strategic goals (defined in the Impact Map) into actionable development tasks, our team employed **Story Mapping**. This agile visualization framework allowed us to frame the product backlog entirely from the perspective of the users' daily journey, ensuring we build a cohesive, end-to-end experience rather than a disjointed set of



features.



Figure 4. Complete User Story Mapping

6. MVP And Release Plan

6.1 Minimum Viable Product (MVP) User Stories

The following user stories define the core functionalities required for the MVP. These stories are prioritized based on their business value and impact on solving the primary user problem.

User Story 1 – Basic Communication

As an autistic child,
I want to select visual icons to form simple sentences,
so that I can express my needs clearly.

- **Business Value:** High
- **Story Points:** 8



User Story 2 – Express Basic Needs

As an autistic child,
I want to quickly indicate common needs
(e.g., food, drink, rest),
so that I can communicate without frustration.

- **Business Value:** High
- **Story Points:** 5

User Story 3 – View Daily Tasks

As an autistic child,
I want to see a simple visual schedule of my day,
so that I understand what will happen next.

- **Business Value:** High
- **Story Points:** 5

User Story 4 – Simple Task Completion

As an autistic child,
I want to mark tasks as completed,
so that I feel a sense of progress and structure.

- **Business Value:** Medium
- **Story Points:** 3

User Story 5 – Parent Basic Interface

As a parent,
I want to access a simple interface,
so that I can support my child's communication and schedule.

- **Business Value:** Medium
- **Story Points:** 3



Summary

The execution strategy ensures that the MVP delivers immediate value by focusing on communication and basic structure. Subsequent releases build upon this foundation by adding behavioral tracking and analytical insights, aligning closely with user needs identified during interviews.

6.2 Phased Release Plan

To ensure an efficient and user-centered development process, the project follows a phased release strategy. This approach prioritizes core functionalities first and progressively introduces advanced features based on user needs and feedback.

Release 1 – MVP (Minimum Viable Product)

Objective:

Enable basic communication and essential daily support.

Features included:

Digital PECS Sentence Builder

Basic Visual Schedule (simple daily tasks)

Simple Parent Interface

Rationale:

Communication is the **most critical need** identified in the interviews. Without it, the core problem is not solved. Therefore, the first release focuses on enabling the child to express basic needs.

Release 2 – Enhanced Support



Objective:

Improve daily structure and begin behavioral tracking.

Features included:

Advanced Visual Scheduling (step-by-step tasks)

Task completion tracking

Private Behavior Logger (for parents)

Rationale:

After communication is established, the next priority is helping the child follow structured routines and allowing parents to record behavioral observations.

Release 3 – Data & Insights

Objective:

Provide insights and long-term support for parents.

Features included:

Automated Weekly Reports

Emotional trend analysis

Data visualization for parents

Rationale:

Once sufficient data is collected, the system can generate meaningful insights to help parents better understand emotional patterns and triggers.